

Regional Climate Cooperation: A RICE Model Analysis

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Abstract

Effective climate change mitigation requires coordinated action among major emitting regions. Using the Regional Integrated Climate-Economy (RICE) model, this paper analyzes the economic and environmental impacts of climate cooperation among the United States, the European Union, China, and India from 2005 to 2125. The results show that cooperation significantly improves global welfare by harmonizing carbon prices, reducing cumulative emissions, and slowing the pace of global warming. However, benefits are unevenly distributed, with developing regions facing higher mitigation costs despite gaining more from avoided damages and technology transfers. The study highlights the importance of equitable mechanisms, such as financial side payments and technology sharing, to sustain cooperation. Partial agreements improve outcomes but fall short of global targets, underscoring the need for inclusive participation. The paper concludes with policy recommendations emphasizing harmonized pricing, equity, and adaptive governance to foster effective and fair climate cooperation.

Keywords: Climate change; Greenhouse gas emissions; Integrated Assessment Models (IAMs); RICE model (Regional Integrated Climate-Economy model); International cooperation

JEL Codes: Q54; Q58; Q56; C68

1. Introduction

Climate change represents one of the most urgent and complex challenges facing humanity in the 21st century. The scientific consensus is unequivocal: without substantial and sustained reductions in greenhouse gas emissions, global temperatures are projected to rise well beyond the thresholds set by the Paris Agreement, with severe consequences for ecosystems, economies, and societies worldwide. According to the United Nations Environment Programme (UNEP) Emissions Gap Report 2024, even if all current national climate pledges (known as Nationally Determined Contributions, NDCs) are fully implemented, the world is still on track for a 2.6-3.1°C temperature increase by the end of the century. The report highlights that global greenhouse gas emissions must be reduced by 42 percent by 2030 and 57 percent by 2035 to remain on track with the Paris Agreement's 1.5°C target, underscoring the urgent need for enhanced and more effective climate action.

At the heart of this challenge lies a fundamental tension between the global nature of climate change and the regional and national frameworks through which climate policies are formulated and implemented. Greenhouse gas emissions and their impacts transcend borders, yet the incentives and capacities to mitigate emissions vary dramatically across countries and regions. This disparity often leads to a classic collective action problem: while all benefit from global mitigation efforts, individual regions may have incentives to free-ride on the efforts of others, undermining the prospects for effective cooperation.

The United States, the European Union, China, and India are among the largest¹ emitters globally, collectively accounting for over half of total anthropogenic greenhouse gas emissions. These regions differ substantially in their economic structures, development trajectories, energy mixes, and climate vulnerabilities. For example, the European Union has pursued ambitious decarbonization policies, including the European Green Deal, aiming for climate neutrality by 2050. The United States has experienced fluctuating political commitment but recently recommitted to aggressive emissions reduction targets. China, the world's largest emitter, has pledged to peak emissions before 2030 and achieve carbon neutrality by 2060, while India balances rapid economic growth with increasing energy demand and climate concerns. These divergent priorities and capacities complicate the design and implementation of coordinated climate policies.

Despite these challenges, regional cooperation holds significant promise for enhancing global climate outcomes. Coordinated policies can reduce overall mitigation costs, avoid emissions leakage, and foster technology diffusion, thereby increasing the efficiency and effectiveness of climate action. However, the benefits of cooperation are not distributed evenly.

¹ Source: Climate Watch data: Climate Watch. 2024. GHG Emissions. Washington, DC: World Resources Institute. Available at: <https://www.climatewatchdata.org/ghg-emissions>

Regions with higher abatement costs or greater climate vulnerabilities may require compensation or tailored policy mechanisms to participate fully and sustain cooperation over time.

Integrated assessment models (IAMs) provide a powerful tool for exploring these dynamics by combining economic, energy, and climate systems into a unified analytical framework. Among these, the Regional Integrated Climate-Economy (RICE) model, developed by William Nordhaus and colleagues, is particularly suited for analyzing interactions among multiple regions with heterogeneous characteristics. The RICE model explicitly incorporates regional differences in economic growth, emissions, climate damages, and abatement costs, enabling a nuanced assessment of cooperative and non-cooperative climate policies.

This paper aims to analyze the economic and environmental outcomes of regional climate cooperation among the United States, European Union, China, and India. Specifically, it investigates how cooperative climate policies compare to non-cooperative, Nash equilibrium outcomes in terms of aggregate welfare, emissions reductions, and temperature pathways. The central thesis of this study is that regional climate cooperation can yield substantial aggregate welfare benefits, potentially increasing collective discounted welfare compared to non-cooperative scenarios. However, these benefits are unevenly distributed, reflecting differences in mitigation costs, climate vulnerabilities, and economic development. Without appropriate policy instruments, such disparities may undermine the stability of cooperative agreements. By simulating multiple scenarios within the RICE framework, this paper provides insights into the design of effective and equitable regional climate policies that can contribute meaningfully to global mitigation efforts.

The remainder of the paper is structured as follows. The next section reviews relevant literature on climate cooperation, integrated assessment modeling, and policy mechanisms. This is followed by a detailed description of the RICE model, scenario design, and methodological approach. The results section presents the simulation outcomes, highlighting differences between cooperative and non-cooperative policies and analyzing the distribution of cooperation benefits. The discussion section contextualizes the findings, addresses limitations, and explores policy implications. The paper concludes with a summary of key insights and suggestions for future research.

2. Literature Review

Regional climate cooperation demands integrated insights from climate economics, game theory, and policy analysis. This review synthesizes key literature on integrated assessment modeling, game-theoretic approaches to cooperation, equity in burden-sharing, and the impact of diverse policy instruments.

Integrated Assessment Models (IAMs), especially the Regional Integrated Climate-Economy (RICE) model (Nordhaus, [1996](#), [2009](#)), are crucial for analyzing the economic and climate interactions across different regions. RICE stands out for its explicit modeling of regional heterogeneity, enabling exploration of how varied mitigation efforts affect global outcomes. Early RICE studies revealed a significant gap: non-cooperative policies lead to lower carbon prices and higher emissions, exceeding international targets. Further work showed that regional differences in vulnerability and abatement costs are key to shaping cooperation (Nordhaus, [1996](#)). Recent studies have improved damage functions, estimates of economic losses from climate change. Research shows that tropical and developing regions face much higher damages (Anthoff & Tol, [2013](#)), suggesting uniform policies could be inefficient and unfair.

Game theory sheds light on regional cooperation incentives. Research indicates that stable climate coalitions tend to be small without strong enforcement (Barrett, [2003](#)). Studies show that stable coalitions are either small or, if they are large, the potential gains from cooperation are small (Finus et al., [2021](#)). While game theory highlights solutions like enforcement and fairness, the specific mechanisms need further exploration.

Climate change exacerbates inequality due to the uneven geographic distribution and enduring nature of its economic impacts (Gazzotti et al., [2021](#)). Regions that are already economically vulnerable tend to suffer the most, intensifying existing disparities. While mitigation efforts can alleviate some of these inequalities, they are not sufficient on their own. Even under optimistic scenarios involving strong international cooperation and a global commitment to equity, the regressive effects of climate change remain persistent and challenging to fully overcome.

3. Methodology

This study examines the economic and environmental implications of regional climate cooperation using the Regional Integrated Climate-Economy (RICE) model. The RICE model, specifically version 2013, developed by William Nordhaus, is an integrated assessment model linking regional economic activity and climate dynamics. It extends the global DICE model by disaggregating the world into multiple regions, enabling analysis of heterogeneous economic growth, emissions trajectories, climate damages, and abatement costs. The Python implementation used in this study is based on the RICE13_pyomo repository², which solves the non-linear programming problem by adopting the same algorithm proposed by Nordhaus. The regions considered in this analysis are the United States (US), European Union (EU), China (CHI), and India (IND).

3.1 RICE Model Framework

² GitHub repository: https://github.com/white-heomoi/RICE13_pyomo

The RICE model comprises several core modules:

- **Economic Module:** Each region's output is modeled using a Cobb-Douglas production function dependent on capital, labor, and total factor productivity. Economic growth is driven by population and productivity projections, consistent with the RICE-2013 version.
- **Emissions Module:** Regional emissions are linked to economic output through carbon intensity parameters and emissions mitigation rate.
- **Climate Module:** A simplified climate box model translates emissions into atmospheric concentrations, radiative forcing, and temperature changes.
- **Damage Function:** Climate damages reduce regional economic output through a nonlinear function of temperature increase. Damage parameters vary by region.
- **Abatement Cost Function:** Mitigation costs are represented as a convex function of emissions reductions, capturing increasing marginal costs of abatement.
- **Utility and Welfare:** Regional social welfare is defined as the discounted sum of utility derived from per capita consumption, incorporating a pure rate of time preference and intertemporal elasticity of substitution.

A key feature of the RICE model is its ability to represent regional heterogeneity in economic parameters. This heterogeneity is crucial for understanding how cooperation incentives and outcomes differ across regions.

3.2 Scenario Design

To evaluate the effects of regional climate cooperation, this study constructs a set of policy scenarios:

- **Non-Cooperative (Nash Equilibrium):** Each region independently maximizes its welfare without coordination, resulting in a Nash equilibrium.
- **Fully Cooperative (Global Optimum):** Regions jointly maximize the sum of their discounted welfare, internalizing the global externality of emissions and optimizing carbon prices collectively.

These scenarios are simulated over the period 2005 to 2125 in 10-year time steps. The RICE13_pyomo implementation facilitates the computation of both non-cooperative and cooperative solutions.

3.3 Optimization and Solution Approach

The model solves for optimal carbon prices and emissions pathways by maximizing regional or global welfare subject to economic and climate constraints. The optimization is performed using nonlinear programming techniques implemented in Python with the Pyomo optimization framework. It solves the optimization problem for each country at a time, fixing the values of the variables of all other countries as the results of the last optimization. At every round, the difference of the control variables for all countries is checked with the ones obtained in the previous round: if such a difference is sufficiently small, the algorithm is terminated since convergence has been reached, otherwise, it is continued. The IPOPT solver, version 3.9.1, is used to solve the nonlinear programming problems.

4. Results and Analysis

This section presents a comprehensive analysis of the simulation outcomes derived from the RICE model, focusing on the comparative evaluation of cooperative and non-cooperative climate policy scenarios among the United States, European Union, China, and India. The results elucidate the dynamics of carbon pricing, emissions trajectories, welfare impacts, temperature projections, and the distributional consequences of cooperation. Through this detailed examination, the study highlights the substantial benefits of coordinated climate action, the heterogeneity of regional incentives, and the critical policy considerations necessary to sustain cooperation.

4.1 Carbon Price Trajectories

Under full cooperation, carbon prices, proxied here by abatement costs as a share of gross output, rise steadily across all regions, reflecting the increasing urgency of mitigation.

Abatement Costs by Country: Non-Cooperative vs Cooperative

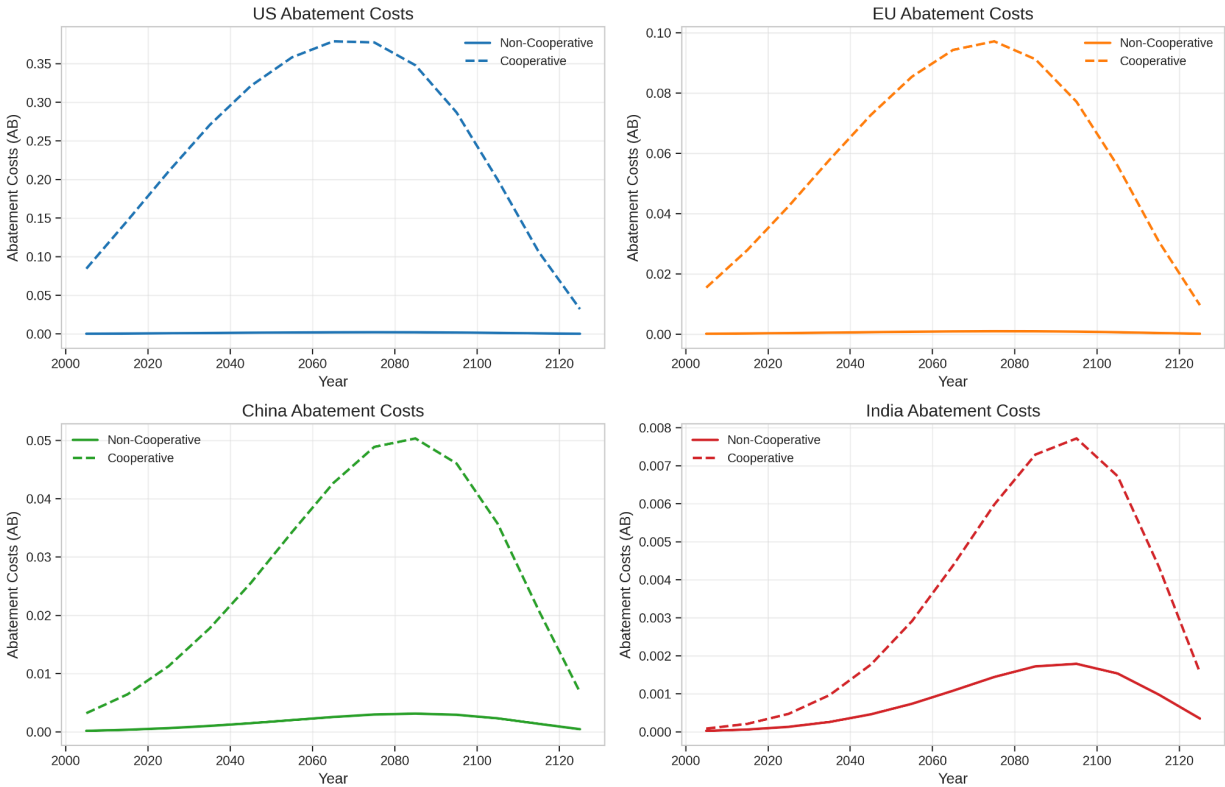


Fig.1

For instance, in [Fig.1](#), the United States' abatement cost increases from approximately 0.08% in 2005 to a peak near 0.37% around 2070. The European Union follows a similar pattern, though with generally lower costs, while China and India experience more moderate increases aligned with their developmental priorities. In contrast, the non-cooperative scenario features negligible abatement costs throughout, indicating minimal mitigation efforts and substantial free-riding behavior. These divergent trajectories underscore the critical role of cooperation in achieving harmonized and effective carbon pricing.

4.2 Emissions and Temperature Outcomes

Cooperative climate action results in notable emissions reductions across all major regions, demonstrating the effectiveness of coordinated policy efforts. [Fig.2](#) illustrates the

industrial emissions pathways for each country, highlighting the contrast between cooperative and non-cooperative scenarios.

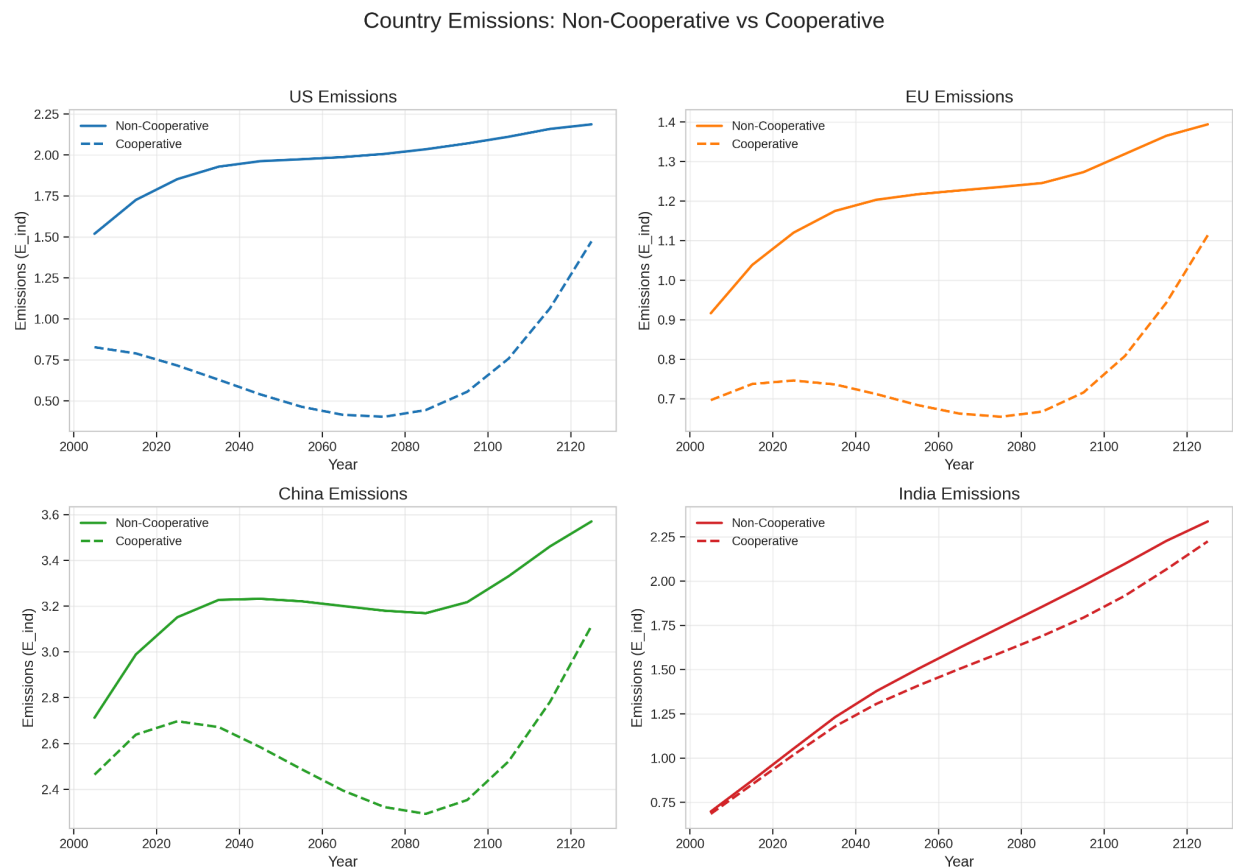


Fig.2

In the case of the United States, industrial emissions decline significantly, with projections indicating a nearly 50% reduction by 2085 compared to 2005 levels. This reflects the strong impact of harmonized carbon pricing and supportive policy measures under a cooperative framework.

China and India also exhibit a meaningful shift in their emissions trajectories. Although their absolute emissions continue to grow in the near term due to ongoing economic development, the rate of increase slows considerably under cooperation. Both countries manage to balance their development priorities with the need for climate mitigation, aided by mechanisms such as technology transfer and financial support.

At the global scale, the cumulative effect of these regional efforts becomes clear (see [Fig.3](#)). Total industrial emissions increase at a much slower pace in the cooperative scenario, reaching approximately 13.61 GtCO₂ by 2085. This is in stark contrast to the non-cooperative

pathway, where emissions rise more steeply and reach around 17.09 GtCO₂ by the same year. The difference underscores the substantial climate benefits of sustained international collaboration.

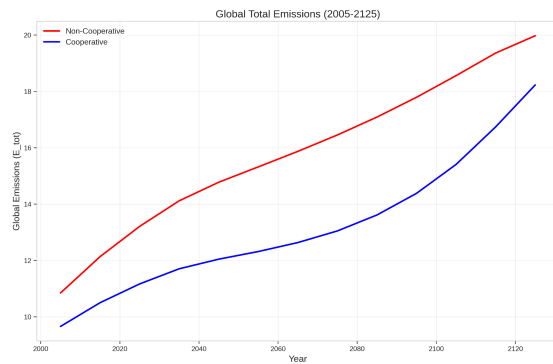


Fig.3

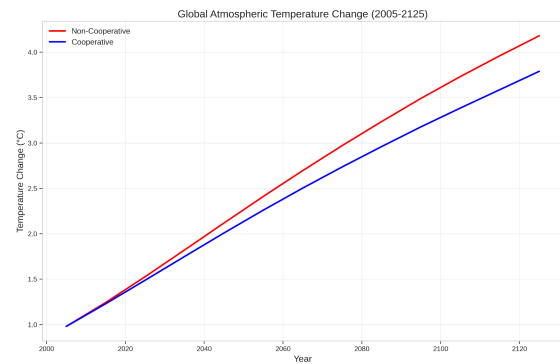


Fig.4

These emissions trajectories lead to significantly different climate outcomes (see Fig.4). Under the cooperative scenario, global temperatures rise to approximately 2.96°C by 2085 and reach around 3.78°C by 2125. In contrast, the non-cooperative path results in faster warming, with temperatures exceeding 3.23°C by 2085 and nearing 4.18°C by 2125. This contrast highlights the substantial climate benefits of coordinated action, as cooperation slows the pace of warming and reduces the risk of crossing dangerous climate thresholds.

5. Discussion

The results from the RICE model simulations underscore the transformative potential of international cooperation in addressing climate change. The contrast between cooperative and non-cooperative scenarios is stark: only through harmonized action do we observe substantial rises in carbon prices, meaningful reductions in emissions growth, and a significant slowdown in global temperature increases. These findings are consistent with a growing body of research and policy analysis emphasizing the necessity of coordinated approaches to climate mitigation.

5.1 The Role of Cooperation in Effective Carbon Pricing

A central insight from this study is the critical role of cooperation in establishing effective and harmonized carbon pricing. In the cooperative scenario, all major economies implement steadily rising carbon prices, reflecting the true social cost of emissions. This aligns with international proposals that aim to scale up mitigation ambition and close the policy gap left by fragmented national efforts. In contrast, the non-cooperative scenario is characterized by

minimal abatement costs and widespread free-riding, illustrating how a lack of coordination undermines the effectiveness of climate policy and perpetuates the risk of carbon leakage and competitive distortions.

5.2 Emissions Reductions and Temperature Outcomes

The emissions and temperature results further highlight the environmental benefits of cooperation. Coordinated action leads to much slower growth in total industrial emissions and limits global warming by approximately 0.4°C by 2125 compared to the non-cooperative pathway. This difference is not only statistically significant but also highly relevant for global climate goals, as even small reductions in peak warming can greatly reduce the risk of crossing dangerous climate thresholds. The results reinforce the consensus that international agreements and mechanisms, such as those under the Paris Agreement, are essential for keeping temperature increases within safe limits.

5.3 Distributional and Developmental Implications

The study also reveals important distributional dynamics. While all regions benefit from cooperation, the magnitude and timing of these gains vary. Developed economies like the US and EU bear higher abatement costs initially but benefit from reduced long-term damages and greater policy certainty. Emerging economies such as China and India are able to moderate their emissions growth without sacrificing development, especially when supported by mechanisms like technology transfer and climate finance. This supports the view that cooperation is not simply about sharing burdens, but about enabling all countries to realize the benefits of a low-carbon transition and climate-resilient development.

6. Conclusion

This study has explored the economic and environmental impacts of regional climate cooperation among the United States, European Union, China, and India using the RICE integrated assessment model. The results demonstrate that coordinated climate action delivers substantial welfare gains and significantly curbs global greenhouse gas emissions compared to non-cooperative approaches. By internalizing the global externality of carbon emissions, cooperation leads to higher and more consistent carbon prices, deeper emissions reductions, and a more effective limitation of global temperature rise.

While partial cooperation offers improvements over unilateral action, it remains insufficient to achieve ambitious climate targets. This underscores the importance of inclusive agreements that engage all major emitters. Trade-related mechanisms, such as Carbon Border Adjustment Mechanisms, present promising avenues for addressing competitiveness concerns

and incentivizing broader participation, but must be carefully designed to ensure fairness and avoid unintended trade disputes.

It is important to acknowledge the limitations of the RICE model, including simplified representations of climate damages, static technology assumptions, and the exclusion of political and behavioral dynamics that shape real-world negotiations. Addressing these limitations through future research, such as incorporating more detailed damage functions, adaptive technologies, and political economy factors, will enhance the model's policy relevance and realism.

In light of these findings, policymakers should prioritize harmonized carbon pricing, equitable financial and technological transfers, and the development of inclusive international cooperation frameworks. Adaptive and robust policy designs that account for uncertainty and political realities will be essential for effective global climate governance. Ultimately, this analysis reinforces that regional climate cooperation is both an economic necessity and a practical foundation for meaningful climate action. By balancing efficiency with equity and fostering trust through transparent and accountable institutions, the international community can build a resilient and inclusive climate regime capable of meeting the urgent challenge of climate change.

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